

Serial communication

1.1 Introduction

Please read "0.520 motor introduction and usage" first to understand the motor parameters, wiring method, and power supply voltage you are currently using. To avoid burning the motherboard or motor.

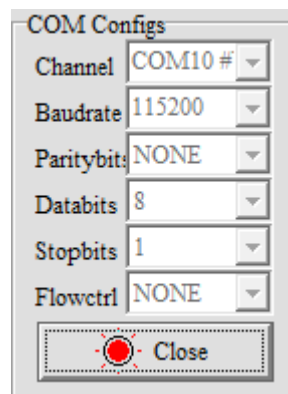
This method is to connect to the computer through the TYPE-C port on the driver board, and use the serial port assistant to send commands to the driver board for control and data reading.

Hardware wiring:

Motor	Four-way motor driver board (Motor)
M2	M-
V	3V3
A	H1A
B	H1B
G	GND
M1	M+

1.2 Instructions

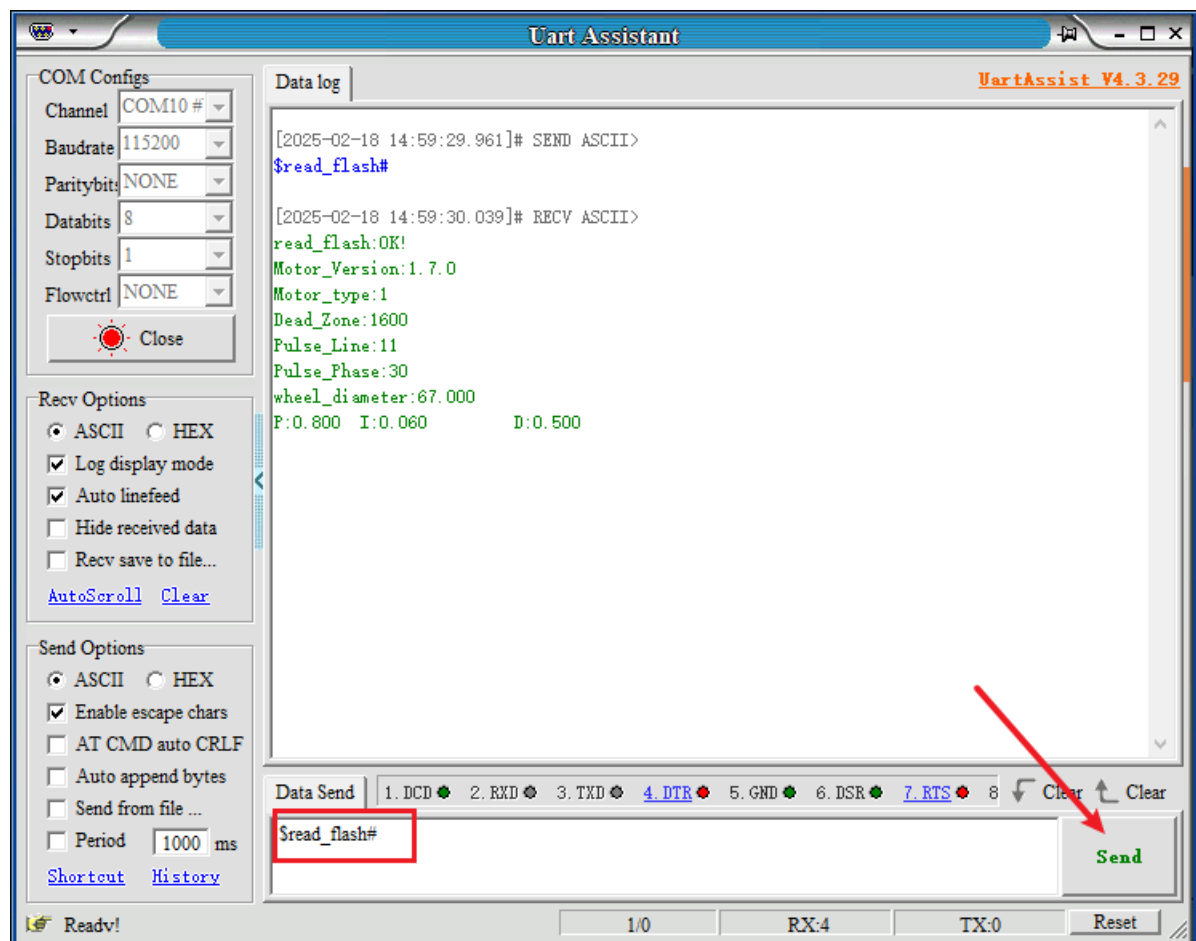
Open the serial port assistant software on the computer, here is Uart Assistant as an example:



Serial port configuration: **Baud rate 115200, no parity check, no hardware flow control, 1 stop bit**

After configuration, you can send instructions in the send window below to control the driver board. All serial port instructions and explanations are listed in the document "4-way motor driver board control instructions".

Next, we will demonstrate how to modify the default parameters to use **310 motor, 48mm diameter tire**.



First, issue the command `$read_flash#`, which is to query the power-off saving parameters in the flash. It can save tire type, motor dead zone, motor phase line, motor reduction ratio, wheel diameter, and motor PID parameters.

Then, according to the parameters of the motor and tire, enter the parameter configuration command in the send column below:

```
[2025-01-21 15:28:20.737]# SEND ASCII>
$mttype:2#
```

```
[2025-01-21 15:28:20.794]# RECV ASCII>
Set mttype:OK!
```

```
[2025-01-21 15:28:22.210]# SEND ASCII>
$deadzone:1250#
```

```
[2025-01-21 15:28:22.263]# RECV ASCII>
Set deadzone:OK!
```

```
[2025-01-21 15:28:34.058]# SEND ASCII>
$mline:13#
```

```
[2025-01-21 15:28:34.189]# RECV ASCII>
Set mline:OK!
```

```
[2025-01-21 15:28:42.225]# SEND ASCII>
$mphase:20#
```

```
Set mphase:OK!
```

```
[2025-01-21 15:28:50.905]# SEND ASCII>  
$wdiameter:48#
```

```
[2025-01-21 15:28:51.044]# RECV ASCII>  
Set wdiameter:OK!
```

```
[2025-01-21 15:29:02.833]# SEND ASCII>  
$read_flash#
```

```
[2025-01-21 15:29:02.893]# RECV ASCII>  
read_flash:OK!  
Motor_Version:1.5.5  
Motor_type:2  
Dead_Zone:1250  
Pulse_Line:13  
Pulse_Phase:20  
wheel_diameter:48.000  
P:0.800 I:0.060 D:0.500
```

You can see that after each configuration command is issued, the driver board will send back an 'OK!' message, indicating that the setting is successful. Finally, use the command to view the flash to see that the parameters you just want to modify have taken effect. The modification method of PID parameters is the same as these, but it generally does not need to be modified, so it will not be demonstrated here.

If you want to control the movement of the motor, send the command `$spd:0,0,0,0#` or `$pwm:0,0,0,0#`. The spd command is used to control motors with encoders, while the pwm command can control motors with or without encoders. Suppose I want to control the 310 motor plugged into Motor1 on the driver board, which has an encoder, and I send `$spd:100,0,0,0#` to control it. To control the motor connected to Motor2, I modify the second value: `$spd:0,100,0,0#`.